

Module4: Blockchain 1.0 [5L]:

Bitcoin blockchain, the challenges, and solutions, proof of work, Proof of stake, alternatives to Bitcoin consensus

Bitcoin blockchain provides a secure, decentralized digital ledger, but faces significant challenges including low scalability (roughly 7 transactions per second), high energy consumption, and regulatory uncertainty. Solutions involve Layer-2 scaling (Lightning Network), Proof-of-Stake adoption, and cross-chain interoperability to improve speed, efficiency, and mainstream adoption.

Key Challenges of Bitcoin Blockchain

- **Scalability & Speed:** Bitcoin's network can only process around 7 transactions per second (TPS), causing congestion and high fees compared to traditional networks like Visa.
- **High Energy Consumption:** The Proof-of-Work (PoW) consensus mechanism requires immense computing power, leading to environmental sustainability concerns.
- **Regulatory Uncertainty:** Diverse global regulations and government crackdowns on cryptocurrency create legal ambiguity for users and businesses.
- **Interoperability:** Bitcoin operates in isolation, making it difficult to interact with other blockchains like Ethereum or Solana.
- **User Experience (UX):** Managing private keys and understanding complex blockchain technology presents a steep learning curve for new users.

Proposed Solutions and Future Trends

- **Layer-2 Solutions (Lightning Network):** Enables instant, low-cost Bitcoin transactions off-chain, reducing network congestion on the main blockchain.
- **Sharding:** A potential (though technically complex) approach to break transactions into smaller datasets processed simultaneously, improving speed.
- **Proof-of-Stake (PoS):** While Bitcoin uses PoW, the broader adoption of PoS protocols by other chains reduces energy usage and increases transaction throughput.
- **Interoperability Bridges:** Cross-chain bridges and protocols (like Wormhole) enable assets and data to move across different blockchain networks.
- **Standardization & Security Audits:** Improved regulatory clarity and better security practices (e.g., smart contract audits) help mitigate security vulnerabilities and legal risks.

Difference between Proof of Work (PoW) and Proof of Stake (PoS) in Blockchain

Proof of Work (PoW) is a consensus mechanism used to secure blockchain networks. The term was introduced by Markus Jakobsson and Ari Juels in 1999 and later became the foundation of Bitcoin's security model. PoW requires miners to solve complex mathematical puzzles. The first miner to solve the puzzle gets the right to add the next block to the blockchain and receives a block reward.

This process ensures that:

- Every block added is verified through computational effort
- Transactions are validated without relying on a third party

- It is extremely difficult for any single miner to control or manipulate the network

Key Features of PoW

- Based on the Hashcash PoW system
- Miners provide proof that they performed computational work before proposing a block
- Solutions are difficult to compute but easy for others to verify
- Limits block creation speed (e.g., Bitcoin creates one block every 10 minutes)
- Reduces the risk of a 51% attack, as gaining majority hash power is extremely expensive
- Builds a trustless, transparent, decentralized network
- Over time, mining can become monopolized by those with powerful hardware

Proof of Stake (PoS)

Proof of Stake is a consensus mechanism where validators are selected based on how much cryptocurrency they lock (stake) in the network. Instead of solving puzzles, validators are chosen algorithmically, making the process far more energy efficient.

In PoS:

- More stake = Higher chance to validate the next block
- Validators earn transaction fees and incentives
- No large-scale mining hardware required

Key Features of PoS

- No complex puzzles or mining hardware, validators stake coins instead
- Validators earn transaction fees, not block rewards
- The first mainstream cryptocurrency to use PoS was Peercoin
- Significantly reduces energy consumption
- Helps prevent monopolies and promotes a more sustainable blockchain model

Proof of Work (PoW) and Proof of Stake (PoS) in blockchain:

S. No	Proof of Work (PoW)	Proof of Stake (PoS)
1	Mining depends on computational power	Validation depends on stake amount
2	Miners receive block rewards	Validators earn transaction fees / staking rewards
3	Requires solving complex puzzles	Block creator selected algorithmically
4	Attack requires majority hash power	Attack requires majority coin ownership
5	High energy consumption	Highly energy efficient
6	Requires specialized mining hardware	Works on standard server hardware
7	Requires investment in mining equipment	Requires investment in a cryptocurrency stake
8	Used in traditional blockchain mining systems	Used in modern energy-efficient blockchain models

Alternatives to Bitcoin's Proof-of-Work (PoW) consensus focus on increasing energy efficiency, speed, and scalability. Key alternatives include Proof-of-Stake (PoS), Delegated Proof-of-Stake (DPoS), Proof-of-Authority (PoA), and Proof-of-Capacity (PoC). These mechanisms reduce computational requirements, offering faster transactions for decentralized networks.

Here, we will discuss various consensus algorithms and how they work.

1. **Proof of Work (PoW):** This consensus algorithm is used to select a miner for the next block generation. Bitcoin uses this PoW consensus algorithm. The central idea behind this algorithm is to solve a complex mathematical puzzle and easily give out a solution. This mathematical puzzle requires a lot of computational power and thus, the node who solves the puzzle as soon as possible gets to mine the next block. For more details on PoW, please read Proof of Work (PoW) Consensus.

2. **Practical Byzantine Fault Tolerance (PBFT):** Please refer to the existing article on practical Byzantine Fault Tolerance(pBFT).
3. **Proof of Stake (PoS):** This is the most common alternative to PoW. Ethereum has shifted from PoW to PoS consensus. In this type of consensus algorithm, instead of investing in expensive hardware to solve a complex puzzle, validators invest in the coins of the system by locking up some of their coins as stakes. After that, all the validators will start validating the blocks. Validators will validate blocks by placing a bet on them if they discover a block that they think can be added to the chain. Based on the actual blocks added in the Blockchain, all the validators get a reward proportionate to their bets, and their stake increase accordingly. In the end, a validator is chosen to generate a new block based on its economic stake in the network. Thus, PoS encourages validators through an incentive mechanism to reach to an agreement.
4. **Delegated Proof of Stake (DPoS):** This is another type of Proof of Stake consensus algorithm. This type of consensus mechanism depends on the basis of the delegation of votes. The users delegate their votes to other users. Whichever user then mines the block will distribute the rewards to the users who delegated to that particular vote. Refer to the article Delegated Proof of Stake for more.
5. **Proof of Burn (PoB):** With PoB, instead of investing in expensive hardware equipment, validators 'burn' coins by sending them to an address from where they are irretrievable. By committing the coins to an unreachable address, validators earn the privilege to mine on the system based on a random selection process. Thus, burning coins here means that validators have a long-term commitment in exchange for their short-term loss. Depending on how the PoB is implemented, miners may burn the native currency of the Blockchain application or the currency of an alternative chain, such as bitcoin. The more coins they burn, the better their chances of being selected to mine the next block. While PoB is an interesting alternative to PoW, the protocol still wastes resources needlessly. And it is also questioned that mining power simply goes to those who are willing to burn more money.
6. **Proof of Capacity:** In the Proof of Capacity consensus, validators are supposed to invest their hard drive space instead of investing in expensive hardware or burning coins. The harder drive space validators have, the better their chances of getting selected for mining the next block and earning the block reward.
7. **Proof of Elapsed Time:** PoET is one of the fairest consensus algorithms which chooses the next block using fair means only. It is widely used in permissioned Blockchain networks. In this algorithm, every validator on the network gets a fair chance to create their own block. All the nodes do so by waiting for a random amount of time, adding proof of their wait in the block. The created blocks are broadcasted to the network for others' consideration. The winner is the validator which has the least timer value in the proof part. The block from the winning validator node gets appended to the Blockchain. There are additional checks in the algorithm to stop nodes from always winning the election, and stop nodes from generating the lowest timer value.

Bitcoin scripting language and its use.

Bitcoin Script is the programming language that underpins Bitcoin transactions. If you're wondering "what is Bitcoin Script?", it serves as a crucial mechanism that allows users to define the conditions under which funds can be spent. Unlike traditional financial transactions, Bitcoin transactions involve the execution of specific conditions encoded in Bitcoin Script.

It operates by manipulating items on a stack, with various opcodes representing operations that can be performed. This scripting language is deliberately limited and designed to be non-Turing complete for security reasons, preventing potentially malicious or infinite loops from being executed on the Bitcoin network.

Bitcoin Script is a **low-level, non-Turing complete language**. This means:

- It is intentionally limited (no loops, no complex logic)
- It is designed for **security and predictability**
- It executes operations using a **stack (LIFO: Last In, First Out)**

Each transaction contains a script that must evaluate to **TRUE** for the transaction to be valid.

Structure of Bitcoin Script

A typical Bitcoin transaction uses two scripts:

1. **ScriptPubKey (Locking Script)**
 - Defined by the sender
 - Specifies the condition to spend the Bitcoin
2. **ScriptSig (Unlocking Script)**
 - Provided by the receiver
 - Supplies the data to satisfy the condition

Key Features

- **Stack-based execution**
- **Deterministic (same input → same output)**
- **Non-Turing complete (no loops)**
- **Supports cryptographic operations**
- **Secure and minimalistic design**

Uses of Bitcoin Script

1. Transaction Validation

Ensures only the rightful owner can spend Bitcoin.

2. Multi-signature Transactions

Requires multiple approvals before spending funds. Example: 2-of-3 signatures needed.

3. Time-locked Transactions

Funds can only be spent after a specific time/block height.

4. Escrow Services

Third-party mediator can help resolve disputes.

5. Pay-to-Script-Hash (P2SH)

Allows complex conditions to be hidden until spending.

6. Smart Contract Basics

Though limited, it enables simple smart contract functionality (compared to platforms like Ethereum).

Limitations

- No loops or recursion
- Limited flexibility compared to full smart contract platforms
- Not suitable for complex decentralized applications (DApps)